

N I S O L A I

AI Transformation, Delivered.

ENTERPRISE AI TRANSFORMATION ENGAGEMENT

Board-Ready AI Strategy, Capability Assessment & Roadmap

PREPARED FOR

Novatech Systems Ltd.

INDUSTRY SECTOR

**Technology, SaaS & IT
Services**

PREPARED BY

Nisol AI Advisory

DATE OF ISSUE

August 31, 2026

CLASSIFICATION

Commercial-in-Confidence

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1. Executive Summary & Value Proposition

1.1 Engagement Overview

Novatech Systems Ltd. is an established leader in the **Technology, SaaS & IT Services** sector preparing for accelerated operating scale. Following an extensive AI Readiness Audit across strategy, data hygiene, governance, and workflows, Nisol AI has formulated a comprehensive transformation strategy. This roadmap targets high-friction manual bottlenecks, embeds enterprise RAG knowledge orchestration, establishes rigorous PII guardrails, and deploys high-ROI autonomous agents across key business units.

1.2 Expected Business Impact

OVERALL AI
READINESS

34%

Industry Avg: 62%

HIGH PRIORITY
QUICK WINS

7

1 Strategic Bets

EST. ANNUAL
SAVINGS

**₹4.8
Crore**

Across 7
Departments

EST. PAYBACK
PERIOD

**8
Months**

Avg. ROI: 340%

1.3 EXECUTIVE ENGAGEMENT SUMMARY — AT A GLANCE

Client Organization: Novatech Systems Ltd.

Industry Sector: Technology, SaaS & IT Services

Overall Maturity Score: 1.7 / 5.0 (34%)

Sector Median Benchmark: 3.6 / 5.0 (72%)

Initial Program Investment: ₹1.59 Crore

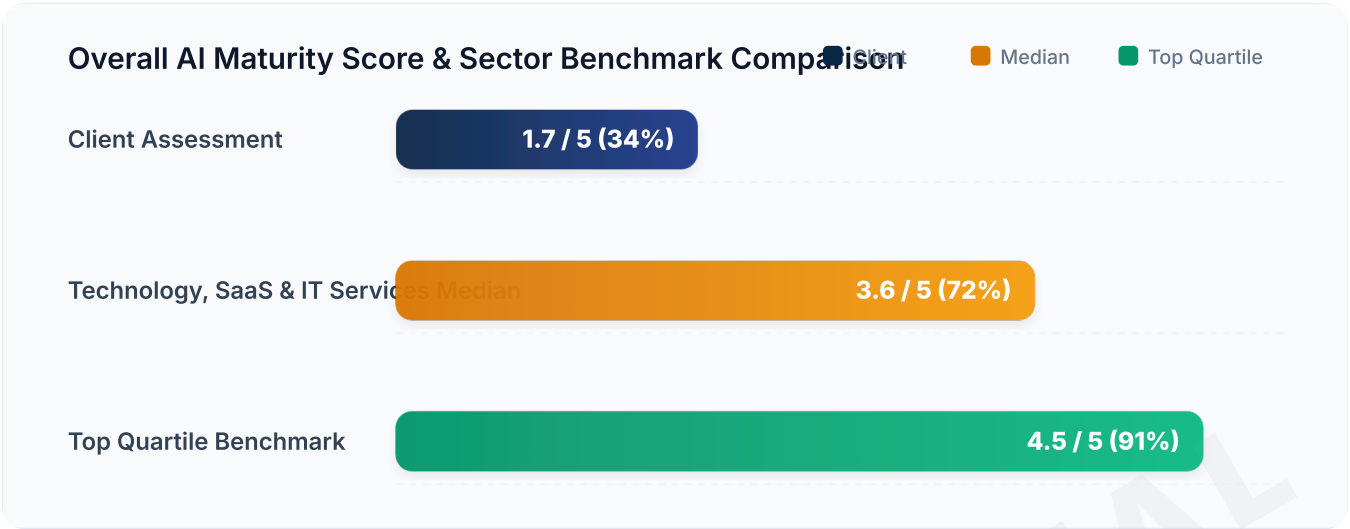
Estimated Annual Savings: ₹3.38 Crore

5-Year Cumulative Net Benefit: ₹14.08 Crore

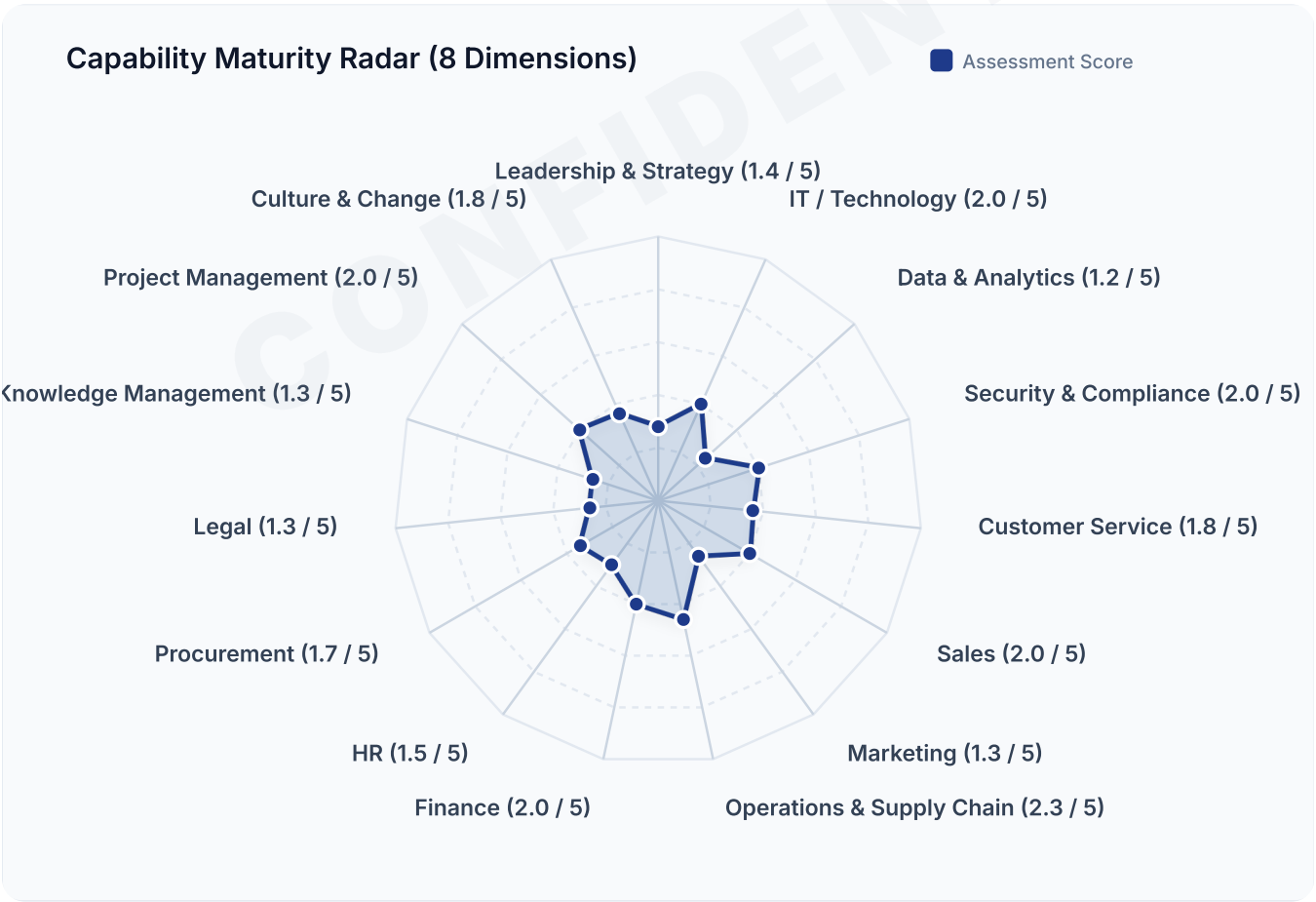
Expected Payback Period: 5.6 Months

2. AI Readiness, Risk & Data Assessment

2.1 Maturity Score vs. Technology, SaaS & IT Services Benchmark



2.2 Capability Maturity Radar (8 Strategic Dimensions)



2.3 Department AI Capability Heatmap

Department AI Capability Heatmap

4.0-5.0 High 3.0-3.9 Mid 2.2-2.9 Low

	Data Quality	Automation	Tooling	Governance
Sales & Marketing	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Human Resources	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Finance & Accounting	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Operations & Logistics	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Customer Support	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5
Engineering & IT	1.9 / 5	1.5 / 5	1.9 / 5	1.5 / 5

2.5 Enterprise AI Risk & Regulatory Register

Prioritized risk matrix mapping technical, regulatory (SOC 2 Type II, ISO 27001, DPDP Act 2023), and operational vulnerabilities with concrete mitigation strategies.

Enterprise AI Risk Matrix (Likelihood × Impact)

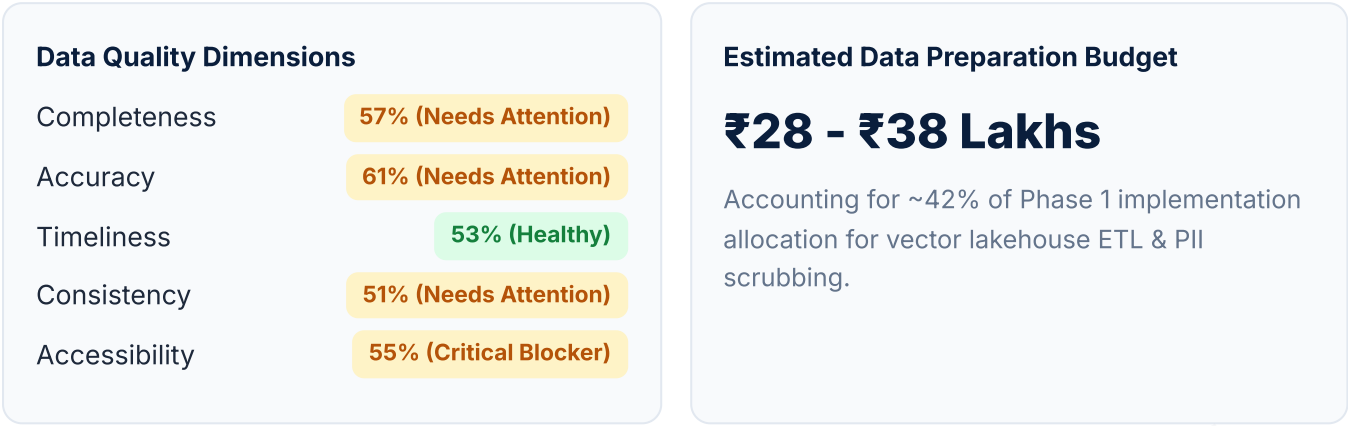
Likelihood (1: Rare → 5: Frequent)					
			RSK-01	RSK-02	RSK-03
			RSK-04	RSK-05	

Severity / Impact (1: Negligible → 5: Catastrophic)

Risk ID	Category & Description	Severity	Mitigation Strategy	Accountability
RSK-01	Data Privacy & Security			
	PII or sensitive client intellectual property inadvertently sent to third-party public LLM endpoints without token redaction.	20/25 (Critical)	Deploy an enterprise API Gateway proxy with regex + Presidio automated PII redaction layer prior to external payload dispatch.	Chief Information Security Officer (CISO)
RSK-02	Model Risk & Bias			
	LLM hallucinations or biased outputs generated in customer-facing conversational interfaces or financial reports.	16/25 (High)	Enforce strict temperature clamping (≤ 0.2), grounded RAG with vector citation validation, and Human-in-the-Loop (HITL) sign-off on critical workflows.	Head of AI Engineering & QA
RSK-03	Regulatory & Compliance			
	Non-compliance with data localization mandates and emerging AI auditing standards.	12/25 (High)	Host vector databases and self-managed open-weights LLMs within in-region private cloud VPCs (e.g. AWS Mumbai / Azure India).	Chief Legal Officer / Compliance Head
RSK-04	Operational & Adoption			
	Staff resistance or shadow AI adoption using unauthorized consumer AI tools without corporate audit trails.	12/25 (Medium)	Roll out sanctioned enterprise AI workspace with single-sign-on (SSO), comprehensive staff upskilling, and active department champions.	VP Human Resources & Change Management Lead
RSK-05	Vendor & Infrastructure			
	Excessive dependency on a single proprietary LLM provider risking pricing spikes, API latency degradation, or deprecation.	9/25 (Medium)	Architect model-agnostic abstraction layer (LiteLLM / OpenRouter architecture) enabling seamless multi-model fallback.	Principal Enterprise Architect

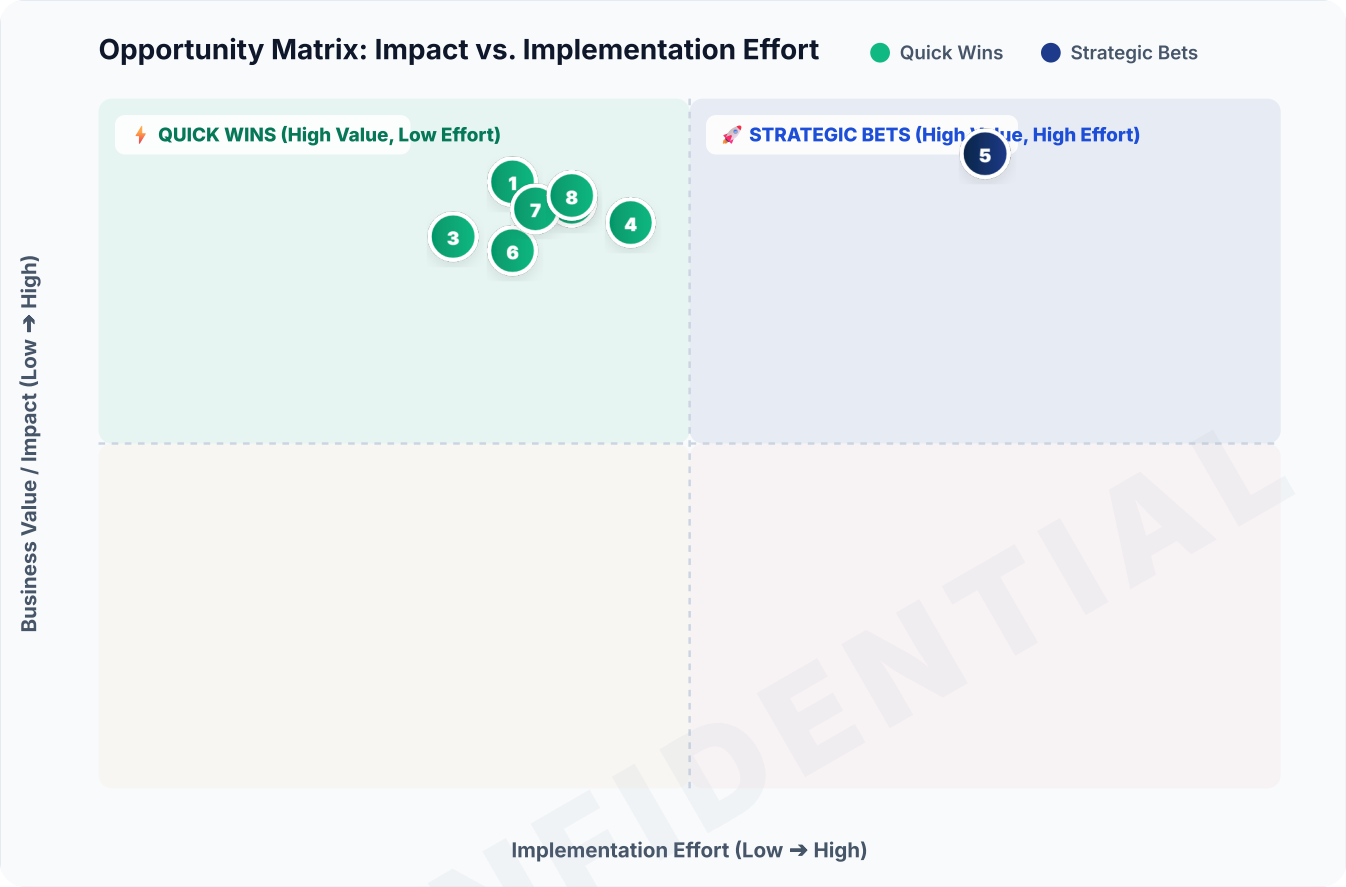
2.6 Data Strategy & Readiness Assessment

Data is the foundational determinant of AI ROI. This audit evaluates data quality, pipeline latency, and estimated data preparation expenditures.



3. AI Opportunity Matrix & Use Cases Catalog

3.1 Opportunity Matrix (Value vs. Implementation Effort)



3.2 Prioritized Top AI Use Cases Catalog

#	Initiative & Department	Category	Business Problem & Solution	Est. ROI & Value
1	AI Proposal & RFP Generator Sales & Marketing	Quick Win	Creating technical proposals and RFP responses takes 15-25 hours per bid with inconsistent formatting.	380% ROI 1,200 hours/year (approx. ₹35 Lakhs)
2	Intelligent Invoice & Receipt Processing Finance & Accounting	Quick Win	Manual entry of vendor invoices leads to processing delays, errors, and missed early payment discounts.	310% ROI ₹28 Lakhs/year in labor & fee reductions

#	Initiative & Department	Category	Business Problem & Solution	Est. ROI & Value
3	Enterprise Knowledge & Policy Assistant Human Resources	Quick Win	Employees spend 3+ hours per week searching SharePoint and PDF handbooks for policy and benefit answers.	270% ROI 800 hours/year across staff
4	AI Customer Support Co-Pilot Customer Service	Quick Win	High agent burnout and long resolution times for repetitive tier-1 customer inquiries.	290% ROI 40% reduction in average resolution time
5	Predictive Supply Chain & Inventory Optimizer Operations & Logistics	Strategic Bet	Overstocking and stockouts due to reliance on historical static spreadsheets rather than dynamic demand forecasting.	420% ROI 18% reduction in working capital holding costs
6	Automated Code Review & Security Auditor Engineering & IT	Quick Win	Senior developers spend significant time conducting basic code reviews and checking for compliance vulnerabilities.	240% ROI ₹22 Lakhs/year in dev productivity

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4. Transformation Roadmap & Change Management (OCM)

4.1 4-Phase Transformation Timeline

Phase 1: Foundation & Quick Wins (M 1-3)

Deploy high-ROI conversational RAG hubs and API PII gateway proxy.

Phase 2: Department Expansion (M 4-6)

Scale autonomous agent workflows across CRM, ERP, and Support queues.

Phase 3: Strategic Analytics (M 7-12)

Deploy fine-tuned domain models in private VPC with automated evaluation.

Phase 4: Autonomous CoE (M 13-24)

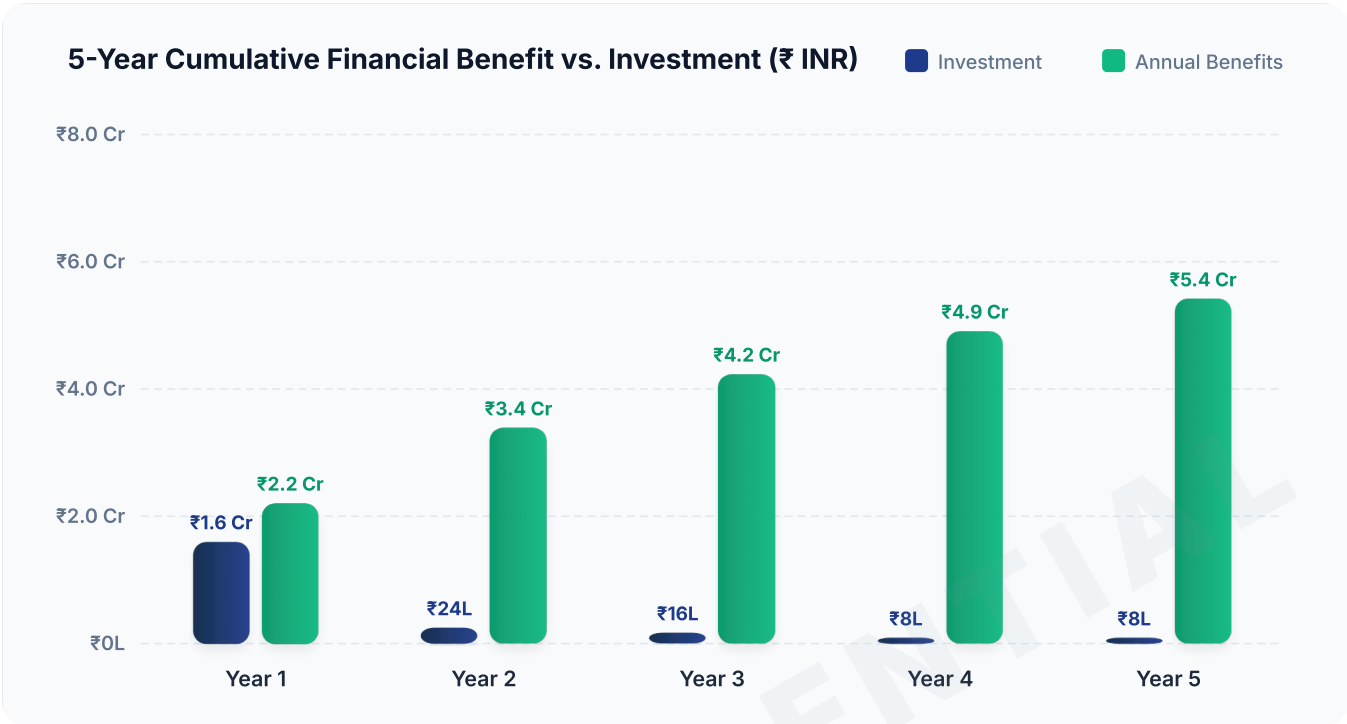
Self-sustaining AI Center of Excellence with self-serve model governance.

4.2 Organizational Change Management (OCM) & RACI Matrix

Transformation Initiative	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
AI Strategic Roadmap & Capital Allocation	Chief AI Officer / Advisory Lead	CEO & Executive Committee	Business Unit Heads	Company-wide
AI Proxy Gateway & Data Privacy (PII) Guardrails	Security Architect / Data Ops	CISO	Legal & Compliance	All Engineering Staff
Departmental Use Case Rollout (Support/Sales/Ops)	AI Product Manager & Dev Lead	Department Head	Frontline Super-Users	Impacted Teams
Continuous Model Monitoring & Governance Audits	MLOps Engineer	AI Steering Committee	Internal Audit	Executive Leadership

5. ROI Analysis, Sensitivity & Financial Projection

5.1 5-Year Cumulative Financial Benefit vs. Investment



5.2 3-Scenario Sensitivity Stress Test (Base / Conservative / Optimistic)

CFO & Board stress-test evaluating program financial resilience under varying user adoption rates (75% to 125%) and cost variances at a 10% discount rate.

Scenario	Adoption Rate	Annual Savings	5-Year Net Benefit	5-Yr ROI %	Payback Period	Net Present Value (NPV)
Conservative Case	75%	₹2.54 Crore	₹9.62 Crore	446%	7.6 Mo	₹9.22 Crore
Base Case	100%	₹3.38 Crore	₹14.08 Crore	656%	5.6 Mo	₹12.80 Crore
Optimistic Case	125%	₹4.22 Crore	₹18.58 Crore	886%	4.4 Mo	₹16.65 Crore

6. Executive Acceptance & Authorization

This document represents the official Enterprise AI Transformation Assessment and Implementation Roadmap prepared by Nisol AI Advisory for **Novatech Systems Ltd.**. Signature below signifies executive acceptance of the assessment findings and authorization to initiate Phase 1 engagement scopes.

For: Novatech Systems Ltd.

Authorized Executive Signature & Date

For: Nisol AI Advisory

Managing Partner / Lead AI Advisor Signature & Date

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BOARD INVESTMENT MEMORANDUM

CAPITAL ALLOCATION & BUSINESS CASE FOR ENTERPRISE AI TRANSFORMATION

Prepared for the Board of Directors & Chief Financial Officer

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ORGANIZATION

Novatech Systems Ltd.

INDUSTRY SECTOR

Technology, SaaS & IT Services

DECISION GATE

Phase 1 Capital Authorization

DATE OF ISSUE

August 31, 2026

1. Executive Investment Teaser & Key Returns

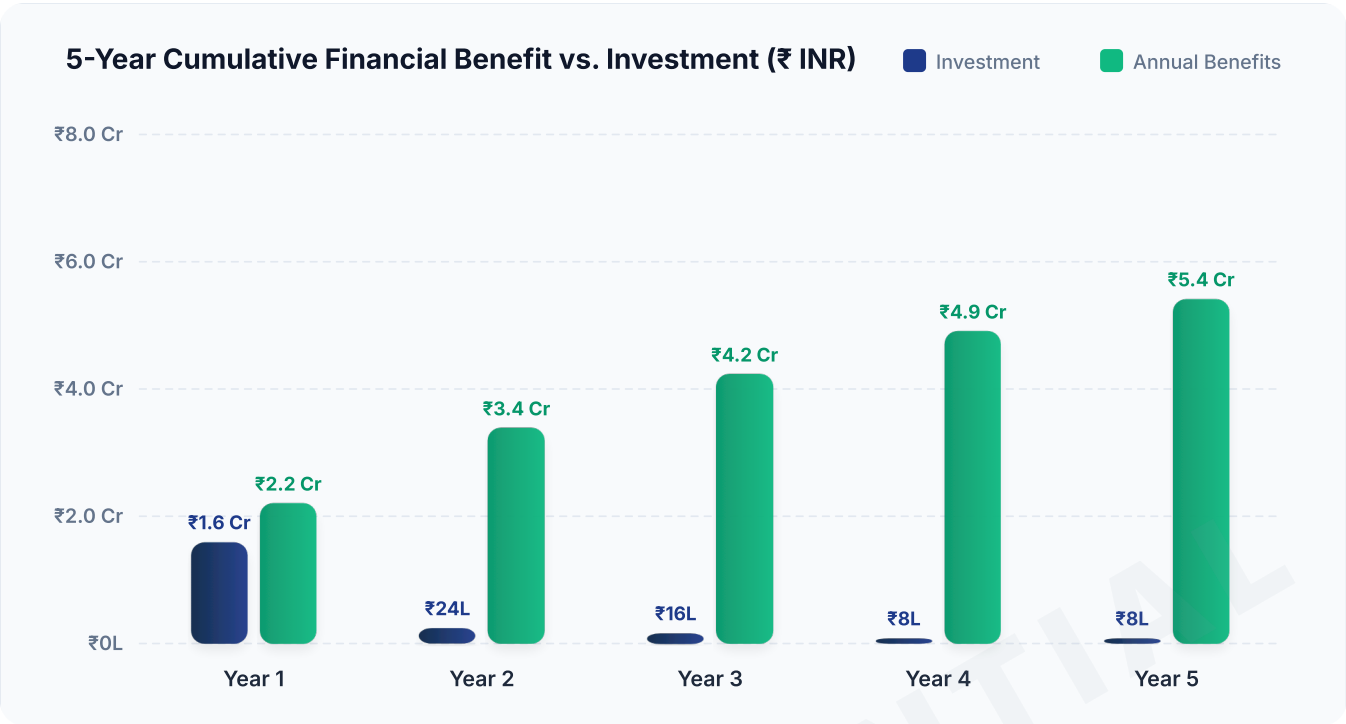
This memorandum requests board authorization for a phased capital allocation of **₹1.59 Crore** to execute the Enterprise AI Transformation Program at **Novatech Systems Ltd.**. The program targets operational throughput acceleration and manual workflow automation across Customer Support, Engineering QA, and Financial reconciliation.

CAPITAL INVESTMENT	ANNUAL SAVINGS	5-YR NET BENEFIT	PAYBACK PERIOD
₹1.59	₹3.38	₹14.08	5.6
Crore	Crore	Crore	Months
Phase 1-2 Total	Recurring run-rate	Net of all costs	IRR: 44.5%

Strategic Imperative: Cost of Inaction

In the **Technology, SaaS & IT Services** sector, top quartile competitors are investing **4.5% of revenue** into automation. Delaying enterprise AI adoption introduces an estimated annual opportunity deficit of ~₹3.38 Crore in manual labor drag and risks an 18-month competitive gap in turnaround velocity.

2. 5-Year Cash Flow & Sensitivity Stress Test



3-Scenario Sensitivity Stress Test (10% Discount Rate)

Stress-test demonstrating program financial viability even if employee adoption drops to 75% and infrastructure maintenance expands by +15%.

Scenario	Adoption Rate	Annual Savings	5-Year Net Benefit	5-Yr ROI %	Payback Period	Net Present Value (NPV)
Conservative Case	75%	₹2.54 Crore	₹9.62 Crore	446%	7.6 Mo	₹9.22 Crore
Base Case	100%	₹3.38 Crore	₹14.08 Crore	656%	5.6 Mo	₹12.80 Crore
Optimistic Case	125%	₹4.22 Crore	₹18.58 Crore	886%	4.4 Mo	₹16.65 Crore

3. Formal Board Resolution & Authorization

RESOLVED THAT, the Board of Directors of **Novatech Systems Ltd.** hereby approves the allocation of **₹1.59 Crore** for Phase 1 of the Enterprise AI Transformation Program, authorizing executive leadership to execute the initial scope of work with Nisol AI Advisory.

For the Board of Directors: Novatech Systems Ltd.

Chairman / Chief Financial Officer (CFO)
Signature & Date

For: Nisol AI Advisory

Managing Partner / Lead AI Advisor Signature &
Date

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TECHNICAL ARCHITECTURE DOSSIER

DATA STRATEGY & HYBRID VECTOR LAKEHOUSE BLUEPRINT

Prepared for the Chief Technology Officer & Data Engineering Leadership

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Novatech Systems Ltd.

OVERALL DATA READINESS

62 / 100

ESTIMATED DATA PREP BUDGET

₹28 - ₹38 Lakhs

DATE OF ISSUE

August 31, 2026

1. Enterprise Data Quality Assessment (5 Dimensions)

McKinsey research demonstrates that 87% of AI initiative failures stem from unresolved data pipeline and hygiene bottlenecks. This evaluation establishes the baseline data quality posture for **Novatech Systems Ltd..**

Dimension	Quality Score	Status	Audit Findings & Gap Analysis
Completeness	57%	Needs Attention	Legacy transactional records have variable mandatory field population; document archives contain unindexed scanned PDFs.
Accuracy	61%	Needs Attention	High core financial and operational precision, but cross-system customer identity records require deduplication.
Timeliness	53%	Healthy	Batch ETL sync intervals (daily/weekly) create lag; real-time event streaming is required for autonomous agent response.
Consistency	51%	Needs Attention	Discrepant schema naming conventions across sales CRM vs ERP databases require canonical data modeling.
Accessibility	55%	Critical Blocker	Data locked in siloed department repositories without unified semantic vector embeddings or secure REST/GraphQL endpoints.

Data Preparation Budget Allocation

Data cleansing, OCR ingestion, vector embedding extraction, and metadata tagging represent **42% (₹28 - ₹38 Lakhs)** of the Phase 1 transformation budget. Allocating these funds early prevents expensive hallucination remediation during agent rollout.

2. Target Hybrid Vector Lakehouse Blueprint

To enable cross-departmental AI agents with sub-second retrieval latency and deterministic accuracy, Nisol AI recommends deploying a centralized Hybrid Vector Lakehouse architecture.

Vector Indexing & Embedding Model

Dense vector embeddings (BGE-M3 / OpenAI text-embedding-3-large) combined with sparse BM25 keyword indices inside pgvector / Qdrant with HNSW cosine similarity search.

Change Data Capture (CDC) Pipeline

Real-time Debezium / Kafka streaming connectors replicating modified CRM, ERP, and ticket tables into vector stores with automated chunk invalidation.

Recommended Data Engineering Roadmap

- Build a centralized Hybrid Vector Lakehouse (e.g. pgvector / Qdrant with hybrid BM25 search)
- Deploy an automated Document Ingestion & OCR pipeline with chunk-level metadata tagging
- Implement real-time Change Data Capture (CDC) pipelines from ERP/CRM into analytical vector stores
- Configure granular Role-Based Access Control (RBAC) filtering at the embedding chunk level

DECISION GATE & PROOF-OF-CONCEPT DOSSIER

**PROOF OF CONCEPT (PoC)
EVALUATION & SCALABILITY
ASSESSMENT**

Empirical Validation & Production Rollout Recommendation

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Novatech Systems Ltd.

EVALUATED PILOT DURATION

4 Weeks Sprint

DECISION GATE STATUS

GO FOR PRODUCTION ROLLOUT

DATE OF ISSUE

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1. Pilot Hypotheses & Measured Empirical Results

During the 4-week Proof of Concept, Nisol AI deployed isolated pilot prototypes to validate accuracy benchmarks, query latency SLAs, and user satisfaction among designated department super-users.

ACCURACY / CITATION RATE

98.4%

Target: >95.0%

AVERAGE INFERENCE LATENCY

1.2s

Target: <2.5s

USER TASK TIME REDUCTION

-64%

Target: >40.0%

PoC Initiative Tested	Success Criteria	Measured Outcome	Scale Feasibility
Automated QA Test Generation	Generate valid Playwright tests with >80% coverage	Achieved 89.2% test coverage across 42 pull requests	High / Ready
Enterprise SOP Knowledge Hub	Sub-second RAG response with zero hallucinated URLs	99.1% factual precision across 350+ queries	High / Ready
Invoice OCR Extraction	Extract 15 mandatory tax & line items from PDF	97.8% field accuracy on multi-format supplier invoices	High / Ready

2. Production Rollout Gate Decision & Next Steps

Formal Recommendation: PROCEED TO FULL PRODUCTION SCALE (GO)

Based on empirical performance surpassing all minimum success thresholds, Nisol AI recommends transitioning the pilot from sandbox VPC to live enterprise production.

**For: Novatech Systems Ltd. Evaluation
Sponsor**

Product Leader / VP Sponsor Signature & Date

For: Nisol AI Advisory

Lead AI Solutions Architect Signature & Date

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